Data Quality Report – Marketing Ads Dataset

🔹 Objective

To identify, validate, and resolve data quality issues in the raw marketing dataset to ensure accurate analysis, reporting, and dashboarding.

1. Issues Identified

1.1 Inconsistent Text Formatting

- Example: `BRAND B`, `Brand B`, `brand b`
- Impact: Incorrect grouping and duplicate insights

1.2 Leading/Trailing Spaces

- Example: `" Brand A"` / `"Brand A "`
- Impact: Treated as different values

1.3 Missing & Null-like Values

- Found values such as:
    - `NaN`, `unknown`, empty strings
- Impact: Breaks aggregations and KPIs

1.4 Data Type Issues

- `date` stored as text
- Numeric columns stored as strings

1.5 Duplicate Records

- Same campaign/ad/date repeated
- Impact: Inflated metrics

1.6 Inconsistent Business Labels

- Example: `fb`, `meta`, `facebook`
- Impact: Fragmented platform analysis

1.7 Invalid Metric Values

- Negative spend/clicks
- Impact: Incorrect business insights

2. Data Cleaning & Handling Strategy

🔹 2.1 Handling Missing Data (VERY IMPORTANT)

Drop Critical Missing Values

```
# Drop records with no date (cannot be used for time-series)
df = df.dropna(subset=['date'])
print(f"Records after date validation: {len(df)}")
```

Reason:

- Date is essential for time-based analysis
- Without date → record becomes unusable

```
# Drop rows where campaign name is missing
df = df.dropna(subset=['campaign_name'])
```

Reason:

- Campaign is a primary identifier
- Missing campaign → cannot map to dimension table

Impute Missing Numeric Values

```
df[numeric_cols] = df[numeric_cols].fillna(0)
```

Reason:

- Missing values in marketing metrics imply:
  → no activity occurred (0 clicks, 0 purchases)
- Prevents errors in aggregation

✅ **Verification**

```
print(df[numeric_cols].isnull().sum())
```

Result:

- Ensured no missing values remain in numeric fields

🔹 2.2 Data Type Correction

- Converted `date` → datetime
- Converted numeric columns using `pd.to_numeric()`

🔹 2.3 Fix Negative Values

```
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col].abs(), errors='coerce')
```

Reason:

- Spend, clicks, impressions cannot be negative
- Used absolute values to correct anomalies

🔹 2.4 Standardization

- Applied `.str.lower()` and `.str.strip()`
- Ensured consistent naming across dataset

🔹 2.5 Metric Recalculation

Recomputed key KPIs to ensure accuracy:

```
df['ctr'] = df['clicks_all'] / df['impressions']
df['cpc'] = df['amount_spent_inr'] / df['clicks_all']
df['cpm'] = (df['amount_spent_inr'] / df['impressions']) * 1000
df['roas'] = df['purchases_conversion_value_inr'] / df['amount_spent_inr']
```

Reason:

- Ensures metrics are derived from clean base data
- Avoids relying on potentially incorrect raw values
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🔍 3. Data Validation Checks

Unique Platform Check

df['data_source_name'].unique()

- Ensured consistent platform naming

Negative Value Check

df[df['amount_spent_inr'] < 0]

- Confirmed no negative spend remains

Missing Critical Fields

df[df['campaign_name'].isnull()]

- Verified no missing campaign values

4. Final Outcome

- Clean and standardized dataset
- Missing values handled appropriately
- Metrics recalculated for accuracy
- Data validated and ready for:
    - PostgreSQL warehouse
    - Power BI dashboards
    - AI-based analysis

📌 Conclusion

A structured data cleaning and validation pipeline was implemented using Python, ensuring high data quality, consistency, and reliability for downstream analytics.

Before:

	Data Source name	Date	Campaign Name	Campaign Effective Status	Ad Set Name	Ad Name	Country Funnel	Geo Location Segment	FB Spent Funnel (INR)	Amount Spent (INR)	...	Checkouts Initiated	Adds of Payment Info	Purchases	Purchases Conversion Value (INR)	Website Contacts	N Conv
0	Brand A	01/02/2026	NaN	PAUSED	TOF | LAL 3-5% | Sales | Womens	Custom | Video 2 | EOSS | 15th Dec 25 – Copy	India	India	0.0	0.0	...	0.0	0.0	0.0	0.0	NaN	
1	Brand A	01/03/2026	Growify | 11th June | TOF | ABO | INDIA | LAL ...	paused	TOF | LAL 3-5% | Sales | Womens	Custom | Video 2 | EOSS | 15th Dec 25 – Copy	India	India	0.0	0.0	...	0.0	0.0	0.0	0.0	NaN	
2	NaN	20/01/2026	Growify | 11th June | TOF | ABO | INDIA | LAL ...	PAUSED	NaN	NaN	INDIA	India	0.0	0.0	...	0.0	0.0	0.0	0.0	NaN	
3	Brand A	01/01/2026	Growify | 11th June | TOF | ABO | INDIA | LAL ...	NaN	TOF | LAL 3-5% | Sales | Womens	Custom | Video 2 | EOSS | 15th Dec 25 – Copy	India	India	0.0	0.0	...	0.0	0.0	0.0	0.0	NaN	

After:

[31]:

	data_source_name	date	campaign_name	campaign_effective_status	ad_set_name	ad_name	country_funnel	geo_location_segment	fb_spent_funnel_inr	amou
1	brand a	01/03/2026	growify | 11th june | tof | abo | india | lal ...	paused	tof | lal 3-5% | sales | womens	custom | video 2 | eoss | 15th dec 25 – copy	india	india	0.0	
2	NaN	20/01/2026	growify | 11th june | tof | abo | india | lal ...	paused	NaN	NaN	india	india	0.0	
3	brand a	01/01/2026	growify | 11th june | tof | abo | india | lal ...	NaN	tof | lal 3-5% | sales | womens	custom | video 2 | eoss | 15th dec 25 – copy	india	india	0.0	
4	brand a	01/11/2026	growify | 11th june | tof | abo | india | lal ...	paused	NaN	custom | video 2 | eoss | 15th dec 25 – copy	NaN	india	0.0	
5	brand a	21/01/2026	growify | 16th june | tof | india | asc | cac ...	paused	tof | asc | cac | india	custom | carousel | eoss | 15th dec25 – copy	india	india	0.0	

2. Data Quality Report - Shopify Dataset

Audit Dimension	Issues Detected	Resolution Strategy
Duplicates	310 redundant rows found.	Removed via full-row matching to prevent spend inflation.
Missing Values	Nulls found in 600+ records.	**Numeric:** Imputed 0 (prevents math errors). **Categorical:** Labeled 'unknown' to keep row visibility.
Integrity	Negative values in Spend and Clicks.	Applied abs() transformation to fix data entry anomalies.
Date Logic	Inconsistent formats and 'NAN' strings.	Standardized to ISO-8601; dropped records missing dates entirely.
Calculated KPIs	Source columns lacked CTR/ROI.	Recalculated from raw metrics using "Safe Division" to avoid #DIV/0 errors.
Categorical	Mixed casing (e.g., 'ACTIVE' vs 'active').	Applied lowercase normalization for accurate SQL GROUP BY operations.